Pacman Game Project Report

Executive Summary

The Pacman Game project is a Java-based implementation of the classic arcade game, developed using object-oriented programming (OOP) concepts in IntelliJ IDEA. This report outlines the project's objectives, design, implementation, and future improvements.

Introduction:

The Pacman Game project aims to recreate the classic gameplay experience, where a player navigates through a maze while avoiding ghosts. This project demonstrates the application of OOP concepts, such as classes, objects, packages, and interaction between system components.

Objectives:

1. Implement OOP concepts in Java
2. Design a simple game structure using packages
3. Simulate player and ghost movement
4. Understand interaction between multiple classes

Tools and Technologies:

- **Programming Language:** Java
- **IDE**: IntelliJ IDEA
- **Version Control:** Git & GitHub

Project Structure:

The project is divided into packages:

1. **main package**
    a. Contains the main game execution file: `Game.java`
2. **person package**
    a. Contains the player (Pacman) class: `Pacman.java`
3. **ghost package**
    a. Contains ghost logic: `Ghost.java`
4. **map package**
    a. Handles the game map and environment: `Gamemap.java`

System Design:

<u>Main Components</u>

1. **Game Class**
    - Controls the game loop

- Starts and runs the game

2. **Pacman Class**
   - Represents the player
   - Handles movement and position
3. **Ghost Class**
   - Represents enemies
   - Moves randomly in the map
4. **Gamemap Class**
   - Stores map layout
   - Checks valid movement positions

Working of the System:

1. The game starts from the `Game` class
2. A map is created using `Gamemap`
3. Pacman is initialized at a starting position
4. Ghosts are created and placed on the map
5. Game loop runs:
   a) Pacman moves based on input
   b) Ghosts move randomly
   c) Collisions are checked
6. Game continues until a stopping condition

Key Features:

1. Modular design using packages
2. Random ghost movement using `Random` class
3. Clear separation of responsibilities
4. Simple and readable code structure

Concepts Used:

1. Classes and Objects
2. Encapsulation
3. Packages
4. Random number generation
5. Basic game loop logic

Challenges Faced:

1. Managing multiple classes and packages
2. Handling movement logic
3. Ensuring correct interaction between objects

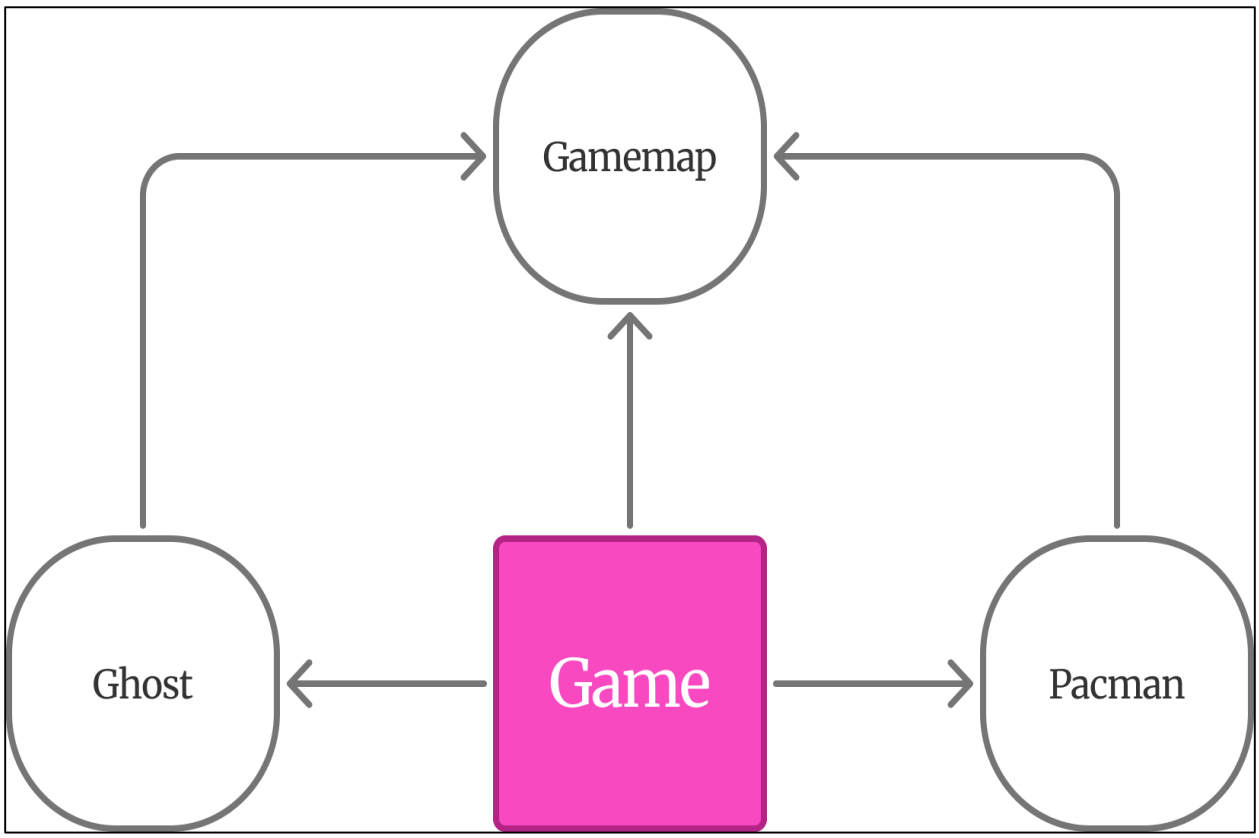
Future Improvements:

1. Add GUI using Java Swing or JavaFX
2. Implement scoring system
3. Add smarter ghost AI
4. Add levels and difficulty increase

Conclusion:

This project successfully demonstrates OOP concepts in building a simple game, providing a foundation for more advanced game development and improving understanding of Java programming and project structuring.

References

- Java Documentation
- IntelliJ IDEA Documentation
- Online programming resources

Architectural Diagram